PATIENT: Male infant born at 33 weeks 4 days gestational age to a 28-year-old G1P0 mother via emergent Cesarean section secondary to placental abruption and fetal bradycardia.

BIRTH WEIGHT: 1,840 grams (appropriate for gestational age). APGAR scores: 6 at 1 minute, 8 at 5 minutes.

PHYSICAL EXAMINATION ON NICU ADMISSION:

General: Preterm infant in moderate respiratory distress with tachypnea, prominent intercostal retractions, nasal flaring, and audible expiratory grunting.

Vital Signs: Heart rate 158 bpm, respiratory rate 72 breaths per minute, BP 54/32 mmHg, SpO2 86% on room air.

Cardiovascular: Regular tachycardia, normal S1 and S2, no heart murmurs. Femoral pulses 2+ bilaterally, capillary refill time 2 seconds.

Respiratory: Bilateral reduced air entry with diffuse fine end-inspiratory crackles throughout both lung fields.

Abdomen: Soft, non-distended, umbilical catheter lines in place. Liver 1 cm below right costal margin.

Neurology: Sucking reflex weak, Moro reflex incomplete, hypotonic posturing typical of 33-week prematurity.

DIAGNOSTIC WORKUP:

Arterial Blood Gas (via umbilical artery catheter): pH 7.24, PaCO2 56 mmHg, PaO2 52 mmHg, HCO3 21 mEq/L, Base Excess -5.2 mEq/L (respiratory acidosis with hypoxemia).

Chest Radiograph: Demonstrates low lung volumes (6 rib expansion), diffuse bilateral reticulogranular 'ground-glass' opacities, and prominent air bronchograms consistent with respiratory distress syndrome (surfactant deficiency).

ASSESSMENT AND PLAN:

1. Neonatal Respiratory Distress Syndrome (RDS) secondary to surfactant deficiency of prematurity.

2. Place on non-invasive continuous positive airway pressure (NCPAP) at 6 cm H2O with FiO2 0.35.

3. Administer intratracheal poractant alfa (Curosurf) surfactant replacement via thin catheter (LISA technique).

4. Maintain intravenous total parenteral nutrition (TPN) with 10% dextrose and amino acids via umbilical venous catheter.

5. Initiate prophylactic intravenous ampicillin and gentamicin pending blood culture results.